studocu

Mid September 2018, questions and answers

Derivative Securities (University of Melbourne)



Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Student Number: | | | | | | | | | | |

THE UNIVERSITY OF MELBOURNE

Department of Finance
Faculty of Economics and Commerce

MID-SEMESTER EXAMINATION – SEMESTER 2, 2018 (Thursday)

SUBJECT NUMBER:	**FNCE 30007**
SUBJECT NAME:	**Derivative Securities**
EXAM DURATION:	**1 hour**
TOTAL MARKS:	**16 marks**
NO. OF PAGES:	**8 (including cover pages)**

Authorised Materials:

1. This is a CLOSED BOOK exam. Students are NOT permitted to have any materials in their possession, except a calculator (but with the restrictions outlined below)
2. Non-programmable calculators are permitted. Calculators with the ability to enter and/or retrieve text are NOT permitted.
3. Students are NOT permitted to use a foreign language dictionary.
4. Students are NOT permitted to take pictures or make any reproduction via any mean of any of the questions in the exam booklet

Instructions to Invigilators:

1. Students should be provided ONLY with this booklet and a multiple choice answer sheet.
2. This booklet and the multiple choice answer sheet MUST be COLLECTED at the end of the exam.

Instructions to Students:

1. Write your student number in the box at the top of this cover page.
2. This exam consists of 16 multiple choice questions. Each question is worth 1 mark.
3. Write your answer to each question IN THE CORRESPONDING BOX ON SIDE 1 OF THE MULTIPLE CHOICE ANSWER SHEET. Make sure you fill in the (appropriate) ovals completely.
4. For each question fill in ONLY ONE OVAL.
5. You may detach the formula list. Do NOT detach any other pages. Examination booklets with missing pages will NOT be marked.
6. Space for rough work is provided below on pages 6 and 7. Examiners will NOT take into account anything written on these pages.
7. This question booklet and the multiple choice answer sheet must be COLLECTED at the end of the exam.

Paper NOT to be held by Baillieu Library.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Maximum possible score: 16 marks

Question 1

A potential *advantage* of futures contracts vs. forward contracts is:

a. The extra liquidity required to cover the potential outflows that occur prior to delivery and caused by daily marking to market
b. Absence of counterparty risk
c. The buyer and the seller can tailor the contract to each other's preferences
d. Longer maturity
e. All of the above

Question 2

Which one of the following IS true:

a. Arbitrageurs in derivatives markets bet on the future direction of price movements: in doing so, they take on large amounts of price risk
b. Hedgers in futures markets always close out their future positions with a loss: the loss is the premium they have to pay in order to get price insurance
c. Speculators in futures markets typically trade futures contracts to reduce the risk of their positions in the underlying assets
d. Futures markets are much less regulated than forward markets
e. None of the above is true

Question 3

A farmer that uses the futures market to lock in the price of wheat is an example of

a. A Long Hedger
b. A Speculator trying to profit from increases in the spot price of wheat
c. A Short Hedger
d. An Arbitrageur trying to exploit price differences between the spot and futures prices of wheat
e. None of the above

Question 4

Assume you are undertaking a short futures hedge. Over the life of the hedge the spot price decreases but the futures price increases by the same amount.

a. Your position is unchanged because you gain in the futures but lose in the spot
b. You are better off because you gain in the futures and in the spot
c. You are worse off because the basis (spot-futures) has increased
d. You are worse off because you lose in the spot and in the futures
e. None of the above statements is true

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 5

Looking at futures contracts for soybeans (a consumption commodity) on March 7 you observe that the futures prices for longer maturities are lower than the futures prices for shorter maturities. You know that interest rates are positive and one incurs costs for storing soybeans. Then, on March 7:

a. The cost of carry for soybeans is negative
b. The convenience yield for soybeans is negative
c. The convenience yield for soybeans is larger than the cost of carry
d. The convenience yield for soybeans is smaller than the cost of carry
e. None of the above statements is true

Question 6

On March 1 the spot price of oil is $60 and the July futures price is $59. On June 1 the spot price of oil is $64 and the July futures price is $63.50. A company entered into a futures contract on March 1 to hedge the purchase of oil on June 1. It closed out its position on June 1. After taking account of the cost of hedging, what is the effective price paid by the company for the oil per barrel?

a. $64
b. $59.50
c. $59
d. $68.50
e. 61.00

Question 7

On May 20, 2016 you entered into a forward contract with your bank to sell 100 ounces of gold, delivery date on November 20, 2016 and delivery price of $1300 per ounce. *Today* is August 20, 2016 and you decide you no longer need the forward agreement and want out of the contract: you observe the following quotes from your bank about the forward prices of gold (per ounce): 1-month $1303.25, 3-month 1309.79, 6-month 1319.65 The risk free rate with continuous compounding is 1% per annum. What happens if you liquidate your forward position today (August 20, 2016)?

a. Pay the bank $1955
b. Receive from the bank $1955
c. Pay the bank $976
d. Receive from the bank $976
e. Pay the bank $1055

Question 8

A company enters into a short futures contract to sell 10,000 units of a commodity for 50 cents per unit. The initial margin is $3,000 and the maintenance margin is $2,500. What is the futures price per unit above which there will be a margin call?

a. 50 cents/unit
b. 45 cents/unit
c. 55 cents/unit
d. 60 cents/unit
e. None of above

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 9

An equity portfolio has a beta of 0.85 and it is currently worth $10,000,000. The fund manager seeks to minimise the risk of the fund over the next 3 months. The 3-month index futures are trading at 5,000 points. The contract size is $25 times the index. The spot return volatility is 1.50, the futures return volatility is 1.6 and the correlation between the index spot and futures returns is 0.92. The fund manager should

a. Short 68 contracts
b. Long 79 contracts
c. Short 54 contracts
d. Long 54 contracts
e. Long 68 contracts

Question 10

A company manufactures jewels and, hence, needs to purchase platinum in the spot market. To hedge price risk he takes a long position in gold futures contracts expiring on November 15. Which one of the following characterizes the company's position:

a. It faces no basis risk if they are certain they can purchase platinum spot on November 15
b. The futures position actually increases the risk of the spot position if the correlation between platinum spot price and gold futures price is not perfect
c. It faces basis risk even if they are certain they can purchase platinum spot on November 15
d. The optimal hedge ratio is equal to 1
e. None of the above statements is correct

Question 11

In the spot FX market the Australian Dollar (AUS) is trading at 1.33 per US dollar (USD). Riskfree interest rates in the United States are 0% while in Australia they are at 2% per annum, with continuous compounding. The futures FX rate for 3-month delivery is $1.34 AUS/USD. Ignoring transaction costs, what should an arbitrageur do?

a. Go long USD futures and short USD spot
b. Go short USD futures and long USD spot
c. Go short both USF futures and USD spot
d. Go long both USD futures and USD spot
e. Do nothing as there is no arbitrage opportunity in the situation described above

Question 12

Which one of the following statements is true:

a. European options have no time value at any time before maturity
b. The intrinsic value of an option can be negative
c. For American options time value is always higher than intrinsic value
d. An out-of-the money put option sells for a non-negative premium because of its time value
e. None of the above statements is true

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 13

A trader reads on his screen that crude oil is trading at $50 per barrel in the spot market and the 6-month risk-free rate is 0%. The trader also knows that, currently, the storage cost of crude oil equals its convenience yield. Which one of the following situations DOES NOT represent an arbitrage opportunity for the trader:

a. A European call option on crude oil with a strike of $40 trading at $52
b. An American put option on crude oil with a strike of $55 trading at $2
c. A 6-month futures contract on crude oil trading at $52
d. An America put option on crude oil with a strike of $50 trading at $55
e. A 6-month futures contract on crude oil trading at $50

Question 14

All other things equal, the writer of a naked call option on NAB shares:

a. Should exercise the option if the spot price of NAB stock climbs above the strike price of the call
b. Faces limited potential gains and unlimited potential losses
c. Has more upside potential than an investor who owns NAB shares
d. Has written the calls thinking that the spot price of NAB stock will go up
e. Faces unlimited potential gains and limited potential losses

Question 15

A one-month European put option on a non-dividend-paying stock is currently selling for $2.50. The stock price is $47, the strike price is $50, and the risk-free interest rate is 6% per annum. Ignoring transaction costs, what should an arbitrageur do?

a. Borrow $49.50 for 1 month at the risk-free rate, short the stock and buy the put
b. Borrow $49.50 for 1 month at the risk-free rate, buy the stock and buy the put
c. Lend $49.50 for 1 month at the risk-free rate, short the stock and write the put
d. Buy the stock, buy the put and exercise it immediately
e. Should do nothing as there are no arbitrage opportunities available in the situation described above

Question 16

Leonard owns Telstra stocks and is thinking of writing a covered call on those stocks. Which one of the following statements IS true:

a. By writing a covered call Leonard seeks protection against increases in the underlying spot price
b. Leonard could receive the same dollar cash flows if he wrote a put on Telstra stocks, with the same strike price and with the same maturity as the call
c. Writing the covered call would be fully equivalent to buying a protective put on Telstra stocks
d. By writing a covered call Leonard seeks protection against declines in the underlying spot price
e. None of the above is true

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

SPACE FOR ROUGH WORK
Examiners will NOT take into account anything written on this page.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

SPACE FOR ROUGH WORK
Examiners will NOT take into account anything written on this page.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

FORMULAE SHEET

Π (short hedge) = $S_t - S_0 - (f_t - f_0)$
Π (long hedge) = $-S_t + S_0 + (f_t - f_0)$

$$b_t = S_t - F_t$$

$$h^* = \frac{\rho_{sf}\sigma_s}{\sigma_f}$$

$$N^* = \frac{h^* Q_A}{Q_F} \qquad N^* = \beta\frac{V_A}{V_F} \qquad N^* = \left(\beta - \beta^*\right)\frac{V_A}{V_F}$$

$$F_0 = S_0 e^{rT}$$
$$F_0 = \left(S_0 - I\right)e^{rT}$$
$$F_0 = S_0 e^{(r-q)T}$$
$$F_0 = S_0 e^{(r-r_f)T}$$
$$F_0 \le \left(S_0 + U\right)e^{rT}$$
$$F_0 \le S_0 e^{(r+u)T}$$
$$F_0 e^{yT} = \left(S_0 + U\right)e^{rT}$$
$$F_0 = S_0 e^{(r+u-y)T}$$

$$F_t = S_t e^{(c-y)(T-t)}$$

$f = (F_0 - K)e^{(-rT)}$ or $f = S_0 - Ke^{-(rT)}$ (long forward)

$f = S_0 - I - K e^{-rT}$ (long forward)

$f = S_0 e^{-qT} - K e^{-rT}$ (long forward)

$C_T = c_T = \text{Max}\{0,\ S_T - K\}$

$P_T = p_T = \text{Max}\{0,\ K - S_T\}$

$c(S_0,T,K) \le S_0$

$C(S_0,T,K) \le S_0$

$p(S_0,T,K) \le Ke^{-rT}$

$P(S_0,T,K) \le K$

$C_t \ge \text{Max}\{0,\ S_t - K\}$

$P_t \ge \text{Max}\{0,\ K - S_t\}$

$$c_t \ge \max\left(S_t - Ke^{-rT}, 0\right)$$
$$p_t \ge \max\left(Ke^{-rT} - S_t, 0\right)$$
$$c + Ke^{-rT} = p + S_0$$
$$c + D + Ke^{-rT} = p + S_0$$

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)